Medical Transformer: Universal Encoder for 3-D Brain MRI Analysis

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***Abstract*— Transfer learning has attracted considerable attention in medical image analysis because of the limited number of annotated 3-D medical datasets available for training data-driven deep learning models in the real world. We propose Medical Transformer, a novel transfer learning framework that effectively models 3-D volumetric images as a sequence of 2-D image slices. To improve the high-level representation in 3-D-form empowering spatial relations, we use a multiview approach that leverages information from three planes of the 3-D volume, while providing parameter-efficient training. For building a source model generally applicable to various tasks, we pretrain the model using self-supervised learning (SSL) for masked encoding vector prediction as a proxy task, using a large-scale normal, healthy brain magnetic resonance imaging (MRI) dataset. Our pretrained model is evaluated on three downstream tasks: 1) brain disease diagnosis; 2) brain age prediction; and 3) brain tumor segmentation, which are widely studied in brain MRI research. Experimental results demonstrate that our Medical Transformer outperforms the state-of-the-art (SOTA) transfer learning methods, efficiently reducing the number of parameters by up to approximately 92% for classification and regression tasks and 97% for segmentation task, and it also achieves good performance in scenarios where only partial training samples are used.**

***Index Terms*— Brain age prediction, brain disease diagnosis, brain tumor segmentation, deep learning, medical image analysis, structural MRI (sMRI), transfer learning, transformer.**

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I. Introduction

MEDICAL image analysis has progressed considerably in the past few years with the use of deep learning [1]. However, building a large 3-D dataset for training deep learning models for medical imaging is challenging, given the difficulty of data acquisition and annotation. The lack of annotated data has prompted the development of transfer learning beyond traditional supervised learning, which enables models to be trained using a small-scale available dataset.

For transfer learning in medical imaging, the standard approach was to pretrain the existing 2-D-based deep neural architectures, e.g., ResNet [2] and DenseNet [3], using large-scale natural image datasets such as ImageNet, and then fine-tune the model on small-sized medical imaging data [4], [5]. This approach yields better performance than the random initial-based training strategy, especially when a small set of samples is available, but the 3-D volume data were split into 2-D slices from three planes and pretrained by 2-D models without recovering the 3-D-form representations, which inevitably caused the loss of the 3-D spatial information. A later study [6] resolved this issue by exploiting the 3-D publicly available models, e.g., recurrent neural networks, trained using the 3-D Kinetics dataset [7].

However, despite having the same 3-D structure, it is worth emphasizing that the medical dataset of interest is very different from the natural images benchmark datasets in computer vision. Medical imaging data focus on specific parts of the body for clinical purposes, resulting in sophisticated yet recurrent patterns. Learning a 3-D medical image network transferred from a natural scene video without considering this characteristic leads to a strong bias in medical image analysis. Thus, it is necessary to create a source model for transfer learning by directly using the medical imaging datasets capable of learning subtle anatomical differences between samples in medical datasets.

To build a source model directly from 3-D medical volume data, recent works have pretrained their models for target tasks such as brain parsing and organ segmentation using labeled datasets [8], [9]. However, these models have been pretrained for specific applications using only a small set of data and are therefore unsuitable to be used as the source model envisioned in this study. Chen et al. [9] built a large-scale dataset from a collection of eight annotated medical datasets and pretrained the 3-D residual network for the segmentation task. Although

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these fully supervised approaches may yield more powerful target models, they demand a substantial amount of annotation efforts to obtain the source model.

Accordingly, many works have adopted self-supervised pretraining to learn image representations from unlabeled medical imaging data [12], [14], [15], [16], [17]. For example, [16], [17] introduced a proxy task for 3-D representation learning by recovering a rearranged and rotated Rubik's cube. Taleb et al. [12] proposed a set of five 3-D self-supervised learning (SSL) tasks that extend the existing 2D SSL tasks to a 3-D-form for medical imaging. More recently, [14] devised an image restoration task from a mixture of transformations to pretrain the model via self-supervision without a labeled dataset. However, despite self-supervision, the existing 3-D-based transfer learning frameworks require too many learnable parameters to be practically applicable to high-dimensional medical datasets.

In this work, we propose a novel transfer learning framework, called *Medical Transformer*, which effectively models a 3-D volumetric MR image as a sequence of 2-D image slices. To make a high-level representation in 3-D-form empowering spatial relations better, we take a multiview approach that leverages plenty of information from three planes of the 3-D volume while providing parameter-efficient training. Specifically, since early studies [18], [19] have already validated the effectiveness of this triplanar representation of 3-D medical images, a given 3-D volumetric image is split into 2-D slices from three orthogonal planes (sagittal, coronal, and axial). Then, these 2-D image slices are fed to the network as inputs. We first pretrain a backbone network consisting of a convolutional encoder and a transformer encoder for masked encoding vector prediction that generates the encoding vectors of the randomly masked images, as our SSL proxy task. Then, after passing through the pretrained backbone network, the 2-D slice features are recovered into 3-D-form representations by their combinations and finally fed into the prediction network for downstream tasks. To validate the effectiveness of our Medical Transformer, we evaluated our pre-trained model on three target tasks that are actively studied in brain magnetic resonance imaging (MRI) research, i.e., brain disease diagnosis, brain age prediction, and brain tumor segmentation, and achieved remarkable performances for all three tasks in comparison to the respective state-of-the-art (SOTA) models.

In summary, the main contributions of this work are as follows.

1) We propose a novel transfer learning framework that takes a multiview approach to effectively model 3-D volumetric images in the form of a sequence of 2-D image slices for high-level volumetric representations.
2) To the best of our knowledge, our study is the first to use the 2.5-D approach and later restore the 3-D-form information for modeling 3-D contextual information while retaining local spatial characteristics in a parameter-efficient manner.
3) We propose a novel backbone network as a source model, which consists of a convolutional encoder and a transformer encoder that extract the hierarchical spatial features of the 2-D slices for each plane, and further model the interslice dependencies via an attention mechanism, thus empowering the representations to reflect global and holistic relations.
4) Upon evaluating our pretrained model on brain disease diagnosis, brain age prediction, and brain tumor segmentation tasks, our Medical Transformer outperformed the SOTA transfer learning methods, efficiently reducing the number of parameters by up to approximately 92% for classification and regression tasks and 97% for segmentation task with relatively low inference times.

II. Related Work

A. Transformers in Vision Models

Many recent works have adopted transformers in vision models. Vision transformer (ViT) [20] divided an image into 16×16 patches and fed these patches into a transformer [21] by treating them as tokens. Despite the simple application of a standard transformer, it showed comparable performance to the convolutional variants by capturing dense and repeatable patterns. However, ViT has a fatal drawback; it has a high computation cost and requires large datasets to outperform the competing convolutional methods.

Our approach is similar to ViT in the sense that we use the transformer as a means of dividing images into subsets and modeling their dependencies. However, our approach takes advantage of the convolutional encoder and the transformer; the convolutional encoder extracts hierarchical spatial features, while the transformer extracts their relational features taking into account interslice dependencies. In addition, the spatial attention of the transformer focuses on important slices, instead of treating each image slice equally, which yields superior performances over 3-D-based competing convolutional methods.

TransUNet [22] used a hybrid CNN-Transformer architecture to exploit both high-resolution spatial information from CNN features and the global context encoded by transformers. This encoding approach of first extracting spatial features with CNN and then modeling the global context with a transformer is very similar to our proposed method. But TransUNet exploited a sequence of flattened 2-D patches as the input and used a 2-D network that processes each 3-D medical image in a slice-by-slice manner. In addition, while TransUNet only focuses on the spatial correlation between tokenized image patches, our method encoded the whole 2-D slice image into slicewise features and modeled their long-range dependencies along the plane axis. Also, another big difference is the decoding part for generating a segmentation mask. TransUNet decoded the hidden features using a cascaded upsampler that consists of multiple upsampling steps via skip connections. In contrast, our method aggregated slicewise features for restoring 3-D volumetric representation, retaining local spatial characteristics.

In addition, Swin UNETR [23] was proposed to use a U-shaped network with a Swin Transformer as encoder and connect it to a CNN-based decoder at five different resolutions by skip connections for the task of multimodal 3-D brain tumor segmentation. Compared with our method, Swin UNETR differs in the configurations of the encoders and decoders.

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TABLE I
SUMMARY OF RELATED WORKS IN TRANSFER LEARNING AND MEDICAL TRANSFORMER (OURS) FOR MODELING APPROACH, MODALITY, SOURCE DATASET, PROXY TASK, AND SUPERVISION

Algorithm	Approach	Modality	Source dataset	Proxy task	Supervision
I3D [7]	3D	Natural	Kinetics [10]	Action recognition on human action videos	Full
NiftyNet [8]	3D	CT	Pancreas-CT [11]	Abdominal segmentation	Full
MedicalNet [9]	3D	CT, MRI	3DSeg-8 [9]	Disease/organ segmentation	Full
3D-SSL [12]	3D	MRI	UK Biobank [13]	3D contrastive predictive coding, 3D rotation prediction, 3D jigsaw puzzles, relative 3D patch location, 3D exemplar networks	Self
Model Genesis [14]	3D	CT	LUNA 2016[1]	Image restoration from four transformations	Self
Medical Transformer	3D Reconstruction from 2.5D	MRI	IXI,[2] Cam-CAN,[3] ABIDE[4]	Triplet loss with intensity transformation, masked encoding vector prediction	Self

[1]https://luna16.grand-challenge.org/, [2]https://brain-development.org/ixi-dataset/, [3]https://www.cam-can.org/, [4]http://fcon_1000.projects.nitrc.org/indi/abide/

Similarly, TransBTS [24] also used transformer for medical image segmentation. TransBTS can model the long-range dependencies in both the slice dimension and spatial dimension simultaneously for volumetric segmentation. While TransUNet relies on pretrained ViT and ResNet models on large-scale natural image datasets, i.e., ImageNet, TransBTS has a flexible network design and is trained from scratch on the task-specific dataset without the dependency on pretrained weights. In contrast, Medical Transformer is well designed for structural MRI (sMRI), relying on pretrained weights from large-scale sMRI datasets, and works well for a variety of sMRI-specific tasks while having a flexible network.

B. 2.5-D Modeling Approaches in Medical Imaging

To handle 3-D medical imaging data, either a 2-D approach is used to learn the 2-D context slice for a specific plane [25], [26] or a 3-D approach is used to directly learn the 3-D volumetric image [27], [28]. However, the former does not model 3-D contextual information well, while the latter involves high computational costs and memory consumption. Accordingly, 2.5-D approaches, e.g., training 2-D networks from three orthogonal planes and finally combining the results, have been proposed. Various methods exist depending on how the results are combined. For example, [29] applied 3-D isotropic Gaussian filtering to propagate the 2-D slice-based probabilities to 3-D by taking local 3-D neighborhoods into account. Roth et al. [18] and Prasoon et al. [19] obtained a combined output by concatenating the outputs of the final layer in three independent CNNs and feeding the result into a softmax classifier. However, these methods have limitations in that these planewise features are not effectively integrated at a higher level; they are simply concatenated without providing volumetric features. In contrast, our proposed 2.5-D method provides volumetric features by combining slicewise features from three planes. This enables modeling 3-D contextual information, retaining local spatial characteristics by fully integrating all the features from three planes. Consequently, it can be applied to various downstream tasks including volumetric tasks such as segmentation.

C. Fully Supervised Pretraining

As there are few annotated datasets in the medical imaging domain, modern approaches have used the pretrained model that is trained using the ImageNet dataset containing over 14 million annotated images [30], [31]. Practically, their pretrained weights from 2-D models, such as ResNet [2] and DenseNet [3], are fine-tuned on medical image analyses, including skin cancer identification [32], Alzheimer's disease (AD) diagnosis [33], and pulmonary embolism detection [34]. However, this strategy has limited applicability for 3-D medical imaging modalities, e.g., computed tomography (CT) and MRI, as it inevitably loses 3-D anatomical information. Accordingly, to capture 3-D spatial information, [6] used inflated 3-D (I3D) [7] trained on the 3-D Kinetics dataset as a feature extractor. However, despite the same 3-D structure, there exists a large domain gap between temporal video data of natural scenes and medical data, which leads to improper modeling of medical contexts. To alleviate these limitations, recent works have tried to pretrain a source model directly using 3-D medical volume data. For example, NiftyNet [8] publicized the pretrained models, i.e., model zoo, for specific applications such as brain parcellation and organ segmentation. However, these models were trained upon a small-scale dataset, making them unsuitable as source models for transfer learning. For training the source model from a large-scale dataset, [9] aggregated eight annotated medical datasets and used them to pretrain a 3-D residual network for a segmentation task.

These aforementioned fully supervised pretraining approaches demand massive, high-quality annotated datasets to obtain the source models for transfer learning. In contrast, our proposed method uses an SSL framework without the need for labeled datasets in 3-D medical image analysis.

D. Self-Supervised Pretraining

SSL has recently gained attention in the field of medical imaging, aiming at learning image representation from unlabeled data [12], [14], [15], [16], [17]. The key challenge for SSL is to define a suitable proxy task from the unlabeled dataset. For example, proxy tasks in computer vision include colorization that recovers a colored image from a gray-scale image [35], [36], image restoration that restores the shuffled small regions within an image to their original position in the image [37], and contrastive learning [38]. In the medical imaging domain, [16], [17] introduced a proxy task of 3-D representation learning by recovering a rearranged and rotated Rubik's cube. Taleb et al. [12] proposed a set of five 3-D self-supervised tasks for medical imaging, namely, contrastive predictive coding, rotation prediction, jigsaw puzzles, relative

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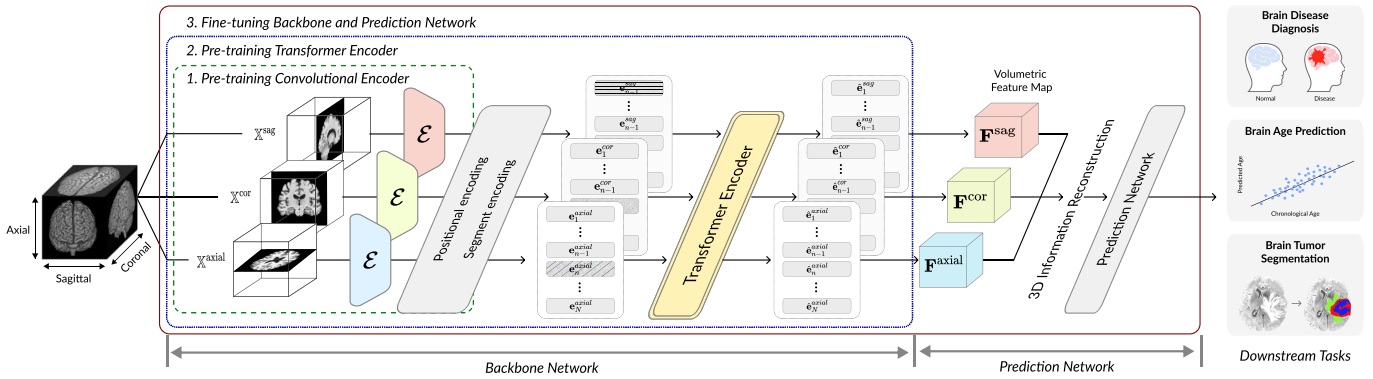


Fig. 1. Schematic of the proposed medical transformer. The double-line of the transformer encoder indicates that the input data of three planes share one transformer encoder, where each plane is distinguished through segment encoding.

patch location, and exemplar networks. Zhou et al. [14] have developed a transfer learning framework, called Model Genesis, to learn general image representation by recovering the original subvolumes of images from their transformed ones derived from nonlinear transformation, local shuffling, outer cutout, and inner cutout.

Our approach differs from the previous self-supervised approaches in obtaining the encoding vector in the representation space, while many related works used an image restoration for a proxy task that recovers the transformed images in the input space. Our masked encoding vector prediction of the proposed method can effectively learn universal representations. In addition, compared with the 3-D-based competing methods that require a huge number of learnable parameters, our method provides a parameter-efficient transfer learning framework and demonstrates superior performances for three medical imaging tasks. We compare the modeling approach, modality, source dataset, and proxy task between the proposed method and the related works in Table I.

III. Proposed Method

The aim of our work is to transfer a backbone network pretrained upon a large-scale 3-D brain MRI dataset to downstream tasks of 3D MRI analyses. In this section, we describe the overall framework of our Medical Transformer, shown in Fig. 1. For learning universal representations of 3-D brain MRI, we first build a large-scale 3-D brain MRI dataset of normal, healthy subjects by collecting three publicly available datasets, i.e., information extraction from images (IXI),[1] Cambridge Centre for Ageing and Neuroscience (Cam-CAN),[2] and Autism Brain Imaging Data Exchange (ABIDE).[3]

Based on a *backbone network* comprising a convolutional encoder and a transformer encoder and a *prediction network*, our Medical Transformer takes a multiview approach, in which a given 3-D volumetric image is split into 2-D slices from three planes (sagittal, coronal, and axial), and these 2-D image slices are fed to the network as inputs. Transfer learning comprises the following three steps: 1) pretraining the convolutional encoder using triplet loss with nonlinear intensity transformation for learning anatomical appearances; 2) pretraining the transformer encoder for a masked encoding vector prediction as our proxy task of SSL for interslice dependencies from slicewise features; and 3) fine-tuning the backbone and prediction network for downstream medical tasks.

A. Notations

In this work, we use a multiview approach that regards a 3-D volume image as 2-D slice images from three planes, i.e., coronal, sagittal, and axial. Given a 3-D volume MRI image $\mathbf{X} \in \mathbb{R}^{w \times d \times h}$, we denote a set of 2-D slice images for each plane, $\mathbb{X}^p = \{X_1^p, \ldots, X_{N_p}^p\}$ ($p \in \{\text{cor}, \text{sag}, \text{axial}\}$), where N_p is the number of slices for the corresponding plane, and w, d, and h present the width, depth, and height of the 3-D volume image, respectively. The 2-D slice images are fed into the convolutional encoder, leading to a set of embedding vectors, $\mathbb{Z}^p = \{\mathbf{z}_1^p, \ldots, \mathbf{z}_{N_p}^p\}$ ($\mathbf{z}_n^p \in \mathbb{R}^{d_{\text{emb}}}$), where d_{emb} denotes the dimension of an embedding vector. Then, we conduct a positional encoding and a segment encoding step before entering the transformer and obtain the encoding vectors, $\mathbb{E}^p = \{\mathbf{e}_1^p, \ldots, \mathbf{e}_{N_p}^p\}$ ($\mathbf{e}_n^p \in \mathbb{R}^{d_{\text{emb}}}$).

B. Medical Transformer Network

Our Medical Transformer consists of a *backbone network* to be pretrained by an SSL proxy task and a *prediction network* based on linear fully connected (FC) layers for final prediction. For the backbone network, we use a 2-D convolutional encoder and a transformer encoder. The convolutional encoder extracts the hierarchical spatial features of the 2-D slices for each plane, and then the transformer encoder extracts their relational features by modeling the interslice dependencies via an attention mechanism, enabling us to capture the dependencies among neighboring and distant slices. In this manner, the representations can reflect global and holistic relations. Here, we use an independent convolutional encoder for each plane that does not share model parameters because the 3D MRI consists of three planes, i.e., axial, coronal, and sagittal, which have different views for each plane. Then, the resulting spatial features go through positional encoding for ordering slices and segment encoding for discerning planes before being fed into the transformer. For segment encoding, with a precomputed embedding matrix from the range corresponding to the number of planes, we made a nontrainable embedding

[1] https://brain-development.org/ixi-dataset/
[2] https://www.cam-can.org/
[3] http://fcon_1000.projects.nitrc.org/indi/abide/

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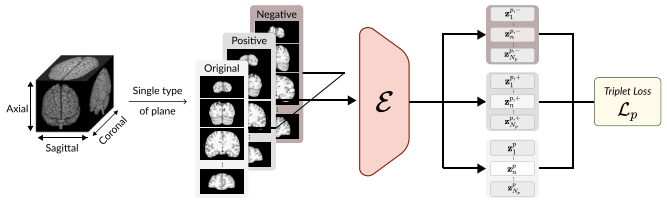


Fig. 2. Illustration of pretraining a convolutional encoder by triplet loss with nonlinear intensity transformation.

layer and conducted an embedding lookup of a range. After passing through the transformer, a volumetric feature map is formed by a combination of slicewise features for all the planes. Our multiview approach of integrating slicewise features from all the planes efficiently reduces the model parameters, still sufficiently capturing plenty of volumetric representations. Finally, the prediction network outputs the final task-specific prediction. Depending on the downstream medical task, we take a multiscale approach to capture low- and high-level features for segmentation and a single-scale approach to capture high-level features for classification and regression.

C. Transfer Learning

1) Pretraining Medical Transformer: To learn a generalized representation using unlabeled data, we first pretrain the convolutional encoder in an SSL manner, as shown in Fig. 2. To this end, we use an exemplar network [39] to derive supervision labels, for which it relies on a nonlinear transformation for learning appearances as one of the image augmentation techniques. The absolute or relative intensity values of medical images deliver crucial information about the image structure. To preserve the intensity of anatomies during image transformation, we use a Bezier curve for a smooth and monotonous transformation, which assigns a unique value to each pixel, ensuring a one-to-one mapping.

For training the exemplar network, we use triplet loss [40], [41], [42], where an original input is compared with positive and negative inputs such that the distance from original to the positive is minimized, and the distance from the original to the negative is maximized. From 3D MRI volume, we introduce original, positive, and negative inputs for three planes, i.e., sagittal, coronal, and axial planes. Specifically, an original input is simply 2-D slice images for a plane. A positive input is defined as the transformed 2-D slice images from the same volume such that their pixel intensity value is nonlinearly one-to-one mapped by the Bezier curve, i.e., intensity transformation only, while a negative input is defined as slice images from other batches (volumes). The original, positive, and negative inputs are fed into the convolutional encoder, and the resulting embedding vectors are average-pooled into $\tilde{\mathbf{z}}^p_{(i)}$, $\tilde{\mathbf{z}}^{p,+}_{(i)}$, and $\tilde{\mathbf{z}}^{p,-}_{(i)}$. The following triplet loss $\mathcal{L}^p_{\text{triplet}}$ is calculated by comparing them with positive pairs and negative pairs as:

$$\mathcal{L}^p_{\text{triplet}} = \frac{1}{N_T}\sum_{i=1}^{N_T} \max\left\{0,\, D\left(\tilde{\mathbf{z}}^p_{(i)}, \tilde{\mathbf{z}}^{p,+}_{(i)}\right) - D\left(\tilde{\mathbf{z}}^p_{(i)}, \tilde{\mathbf{z}}^{p,-}_{(i)}\right) + \alpha\right\} \tag{1}$$

where $\tilde{\mathbf{z}}^p_{(i)}$ is an embedding vector of a random training sample that is averaged over slices, $\tilde{\mathbf{z}}^{p,+}_{(i)}$ is that of the transformed version (positive example), $\tilde{\mathbf{z}}^{p,-}_{(i)}$ is that of a different sample from the dataset (negative example), N_T is the number of training samples, α is the margin between positive and negative samples, and D is a function for distance measure, e.g., $L2$ distance. The final loss is defined by averaging all the losses from the three planes.

Inspired by the successful adoption of masked language modeling as a proxy task of SSL in BERT [43], we propose to learn universal 3D MRI representations for a masked encoding vector prediction as our SSL proxy task. Specifically, we regard a slicewise encoding vector $\mathbf{e}^p_n$ as a token, and a set of encoding vectors $\mathbb{E}^p = \{\mathbf{e}^p_1, \ldots, \mathbf{e}^p_{N_p}\}$, i.e., an image-level feature, as a sentence. We randomly mask out some input slice images for each plane and predict the corresponding encoding vectors of the masked images from encoding vectors of the remaining nonmasked slices. Here, the convolutional encoder pretrained in the previous step is frozen, and only the transformer encoder is trained for the masked encoding vector prediction task so that the transformer encoder can predict the masked encoding vector by determining the interslice dependencies.

For modeling interslice dependencies, we use a standard transformer encoder over slicewise encoding features $E^p \in \mathbb{R}^{N_p \times d_{\text{emb}}}$. The mathematical formulations are as follows:

$$\bar{E}^p = E^p + \text{Softmax}\left(\frac{(W_Q E^p)(W_K E^p)^\top}{\sqrt{d_{\text{emb}}}}\right)(W_V E^p) \tag{2}$$

$$\hat{E}^p = \bar{E}^p + \text{ReLU}(\bar{E}^p F_1) F_2 \tag{3}$$

where $\{W_Q, W_K, W_V\}$ is a set of the learnable weight matrices, and $F_1 \in \mathbb{R}^{d_{\text{emb}} \times d_{\text{ff}}}$ and $F_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{emb}}}$ denote two pointwise feedforward convolutions.

The loss for the masked encoding vector prediction can be formulated using the mean squared error as

$$\mathcal{L}^p_{\text{mask}} = \frac{1}{|\mathbb{M}|}\sum_{m \in \mathbb{M}} \left(\hat{\mathbf{e}}^p_m - \mathbf{z}^p_m\right)^2 \tag{4}$$

where $\mathbb{M}$ is a set of the masked slice indices.

2) Fine-Tuning Medical Transformer: The proposed Medical Transformer aims to build a backbone network that works as a universal MRI encoder, generally transferable for various downstream medical tasks. To validate the effectiveness of our pretrained backbone network, we fine-tune the backbone and prediction network for downstream tasks. To this end, we take two different approaches, i.e., a *single-scale approach* to capture high-level features for classification and regression tasks and a *multiscale approach* to capture both low- and high-level features for a segmentation task. Specifically, the multiscale approach integrates all the outputs from previous convolution layers in the encoder. For instance, as shown in Fig. 3, low- and high-level features from the convolutional blocks, #2–#4 in the encoders $\mathcal{E}^{\{\text{cor,sag,axial}\}}$, are taken into account, which empowers the representations for the segmentation task.

After passing through the backbone network, the encoding vectors $\hat{E}^p \in \mathbb{R}^{N_p \times d_{\text{emb}}}$ for each plane construct a volumetric

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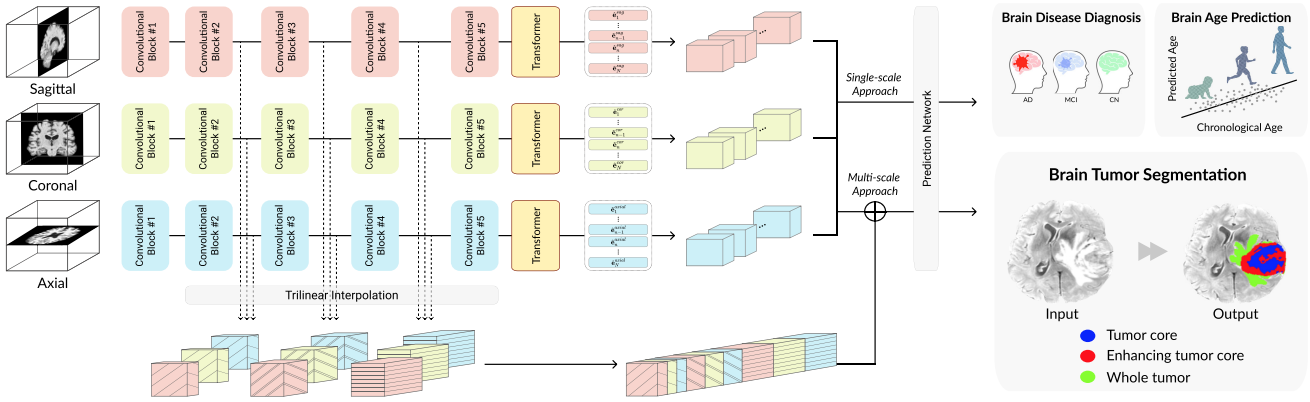


Fig. 3. Illustration of fine-tuning phase for brain disease diagnosis, brain age prediction, and brain tumor segmentation.

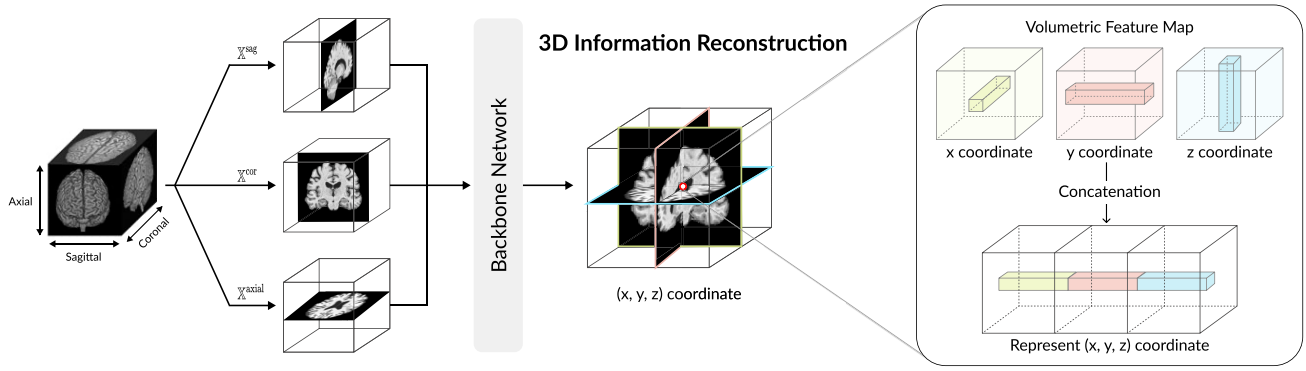


Fig. 4. Illustration of 3-D information reconstruction. One voxel point in the figure considers all the corresponding feature maps from three planes.

feature map $\mathbf{F}^p \in \mathbb{R}^{w\times d\times h\times d_{\text{emb}}}$. In the single-scale approach, slicewise features for all the planes are concatenated into a final volumetric feature map $\mathbf{F} \in \mathbb{R}^{w\times d\times h\times 3d_{\text{emb}}}$ via a combination. That is, one voxel point in the 3D MRI volume considers all the corresponding feature maps from three planes. It should be noted that our proposed method models 3-D data using a 2.5-D method that divides the 3-D data into three planes and then inputs them but restores the 3-D-form information for training the prediction network. As a result, it can model 3-D contextual information, retaining local spatial characteristics, making it suitable for the various downstream tasks including volumetric tasks such as segmentation. This overall process for 3-D information reconstruction is illustrated in Fig. 4.

In the multiscale approach, the feature maps from the preceding convolutional blocks are matched to the same size via trilinear interpolation and additionally concatenated to the aforementioned final feature map, which is fed into the prediction network. The prediction network consists of a linear FC layer for segmentation and a linear FC layer followed by average pooling for classification and regression tasks. During the fine-tuning phase, the pretrained backbone and the prediction network are fine-tuned on downstream tasks.

IV. Experiments

In this section, we evaluated the performance of the proposed Medical Transformer on three medical tasks on publicly available brain MRI datasets. To demonstrate the superiority of our proposed method, we compared the results of three tasks with other SOTA methods in the literature, i.e., I3D [7], NiftyNet [8], MedicalNet [9], and 3-D self-supervised methods (3D-SSL) including 3D-relative patch location (RPL), 3-D-Rotation, 3-D-Jigsaw, 3D-contrastive predictive coding (CPC), and 3-D-Exemplar [12], and Model Genesis [14]. In addition, we conducted extensive ablation studies for our model to evaluate the effects of various components in the proposed method. In addition to quantitative validation, we also compared the qualitative results of a segmentation task. For reproducibility of the results, our pretrained model and code are publicly available at "https://github.com/ku-milab/MedicalTransformer." We specified the complete architecture for pretraining and fine-tuning phase, memory consumption, comparison with 2.5-D competitive methods and SOTA methods of target benchmark dataset, and ablations for our hybrid architecture in Supplementary Material.

A. Experimental Settings

1) Pretraining Medical Transformer: To build a large-scale brain MRI dataset, we collected T1-weighted sMRI scans of 1783 normal, healthy subjects from IXI (#=566), Cam-CAN (#=653), and ABIDE (#=564) datasets. Only subjects who are 13 years of age or older are considered in ABIDE I and ABIDE II. For preprocessing, we conducted Brain ExTraction (BET) [44] to remove areas other than the brain from images (e.g., neck, skull). FMRIB linear image registration toolkit (FLIRT) [45], [46] was used to remove global linear differences over the images such as global translation, scale, and rotation differences. Finally, each 3D MRI volume of $193 \times 229 \times 193$ in size was average-pooled into a size of $96 \times 114 \times 96$ to reduce memory requirements. To handle the data variation problems, i.e., the domain shift caused by

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TABLE II

PERFORMANCE COMPARISON BETWEEN COMPETITIVE METHODS AND THE PROPOSED METHOD FOR BRAIN DISEASE DIAGNOSIS (AD/MCI/NC MULTICLASS CLASSIFICATION), BRAIN AGE PREDICTION (REGRESSION), AND BRAIN TUMOR SEGMENTATION AFTER CROSS-VALIDATION (MEAN ± STD). THE ENTRIES IN BOLD HIGHLIGHT THE BEST PERFORMANCE AMONG THE DIFFERENT APPROACHES FOR THREE TARGET TASKS. (MAUC, MAE, WT, TC, ET)

Pre-training	Approach	Target tasks				
		Classification	Regression	Segmentation		
		mAUC	MAE (years)	Dice WT	Dice TC	Dice ET
No	Training from scratch	0.7728 ± 0.0077	4.4357 ± 0.3260	0.8649 ± 0.0094	0.6392 ± 0.0556	0.4830 ± 0.0485
(Fully) supervised	I3D [7]	0.7325 ± 0.0164	4.6561 ± 0.3209	0.6607 ± 0.0542	0.4708 ± 0.0593	0.0569 ± 0.0159
	NiftyNet [8]	0.5031 ± 0.0165	4.6580 ± 0.3161	0.8395 ± 0.0065	0.5295 ± 0.0148	0.5046 ± 0.0278
	MedicalNet [9]	0.6910 ± 0.0063	4.6443 ± 0.3626	0.7885 ± 0.0378	0.5681 ± 0.0572	0.0809 ± 0.0298
Self-supervised	3D-RPL [12]	0.4849 ± 0.0333	5.1237 ± 0.7086	0.8555 ± 0.0462	0.6595 ± 0.0322	0.3897 ± 0.0078
	3D-Rotation [12]	0.4965 ± 0.0077	4.9799 ± 0.4365	0.8672 ± 0.0344	0.6756 ± 0.0204	0.3717 ± 0.0452
	3D-Jigsaw [12]	0.4950 ± 0.0202	4.7719 ± 0.4784	0.8671 ± 0.0453	0.6739 ± 0.0218	0.3789 ± 0.0415
	3D-CPC [12]	0.4943 ± 0.0109	5.0091 ± 0.7856	0.8879 ± 0.0089	0.6844 ± 0.0086	0.3760 ± 0.0159
	3D-Exemplar [12]	0.5085 ± 0.0163	5.4434 ± 0.9623	$\mathbf{0.8975 \pm 0.0123}$	0.6912 ± 0.0120	0.3819 ± 0.0134
	Model Genesis [14]	0.4997 ± 0.0004	4.6377 ± 0.3411	0.8505 ± 0.0203	0.6201 ± 0.0289	0.0896 ± 0.0329
	Medical Transformer (Ours)	$\mathbf{0.8347 \pm 0.0072}$	$\mathbf{3.4924 \pm 0.0863}$	0.8733 ± 0.0086	$\mathbf{0.6969 \pm 0.0470}$	$\mathbf{0.5882 \pm 0.0437}$

TABLE III

SOCIODEMOGRAPHIC INFORMATION SUMMARY OF ADNI COHORT

Information	CN	MCI	AD
#Subjects (Male)	433 (214)	748 (447)	359 (194)
Age (Years)	74.76 ± 5.82	73.29 ± 7.57	75.29 ± 7.87
Education (Years)	16.29 ± 2.72	15.92 ± 2.86	15.15 ± 3.05
MMSE	29.07 ± 1.12	27.48 ± 1.82	23.21 ± 2.05

data acquisition from multicenters, we conducted spatial and intensity distribution normalization before providing inputs to the model.

The backbone network comprised ResNet-18 and the transformer encoder. The network architecture of the transformer encoder was defined with the dimension of an embedding vector ($d_{\text{emb}} = 16$), the dimension of a feedforward network ($d_{\text{ff}} = 64$), and the number of heads ($h = 4$). For positional encoding, we used a sinusoidal function. We trained our models using the Adam optimizer [47] with an initial learning rate of $1e{-}4$ and a multiplicative decay of 0.99 for 150 epochs (early stopping of 30 patience) using mini-batches of 10 samples.

We divided samples into five-folds, with a single fold for the validation set and the remaining folds for the training set, and chose the final optimal model based on the performance over the validation set. For the masked encoding vector prediction task, the masking ratio was set as 10% so that universal representation was effectively learned by reasonable difficulties, i.e., 9, 11, and 9 slices were masked for the sagittal, coronal, and axial planes, respectively. The pretraining of the Medical Transformer was implemented using PyTorch on GPU NVIDIA GeForce RTX 2080 TI.

2) Downstream Tasks: To validate the effectiveness of our transfer learning framework, we fine-tuned the pretrained model on several medical tasks, i.e., brain disease diagnosis, brain age prediction, and brain tumor segmentation tasks, which are actively studied in brain MRI research. The prediction network was built using a linear FC layer. With respect to the classification and regression tasks, the resulting volumetric map after the linear FC layer was average-pooled. We used Dice loss, cross-entropy loss, and mean squared error loss for segmentation, classification, and regression, respectively.

We reported the performances of our proposed method and competing methods in terms of the average results from the fivefold cross-validation for all the tasks in Table II. Our Medical Transformer and the comparable methods were fine-tuned using PyTorch, and it was trained on GPU NVIDIA GeForce RTX 2080 TI for the classification and regression tasks and TITAN RTX for the segmentation task.

a) Brain disease diagnosis: Our target task is to differentiate between AD, mild cognitive impairment (MCI), and cognitively normal (CN), i.e., multiclass classification, which is a crucial but challenging task in the study of AD, owing to the subtle and diverse morphological changes in the spectra of AD, MCI, and CN [48]. To this end, we used T1-weighted sMRI scans from the AD neuroimaging initiative (ADNI) dataset.[4] The demographic information of the dataset is summarized in Table III. The classification performance was evaluated in terms of the multiclass area under the receiver operating curve (mAUC) [49].

b) Brain age prediction: To verify the versatility of our Medical Transformer on a regression task, we conducted a brain age prediction based on a brain MRI sample, which is regarded as one of the most effective ways for brain aging understanding [50]. T1-weighted sMRI scans from the ADNI dataset were used the same as in the classification task, as presented in Table III. The regression results were reported in terms of the mean absolute error (MAE) between the chronological age and the predicted age.

c) Brain tumor segmentation: To evaluate the performance of the proposed method on brain tumor segmentation, we used the Brain Tumor Segmentation (BraTS) 2020 dataset.[5] It contains 369 training samples and 125 validation samples including T1, postcontrast T1-weighted (T1Gd), T2-weighted (T2), and T2-FLAIR. The dataset was preprocessed with skull-striping, interpolated to a uniform isotropic resolution of 1 mm^3, and registered to an SRI24 space with a dimension of $240 \times 240 \times 155$. The original image size is $240 \times 240 \times 155$, but the largest crop size of $160 \times 192 \times 128$ was used, which ensures that most image content remains within the crop area. As the BraTS dataset consists of four different MRI modalities when compared with a single modality in other datasets, there is a difference in the input channel size of ResNet in the pretrained backbone

[4] http://adni.loni.usc.edu/

[5] https://www.med.upenn.edu/cbica/brats2020/data.html

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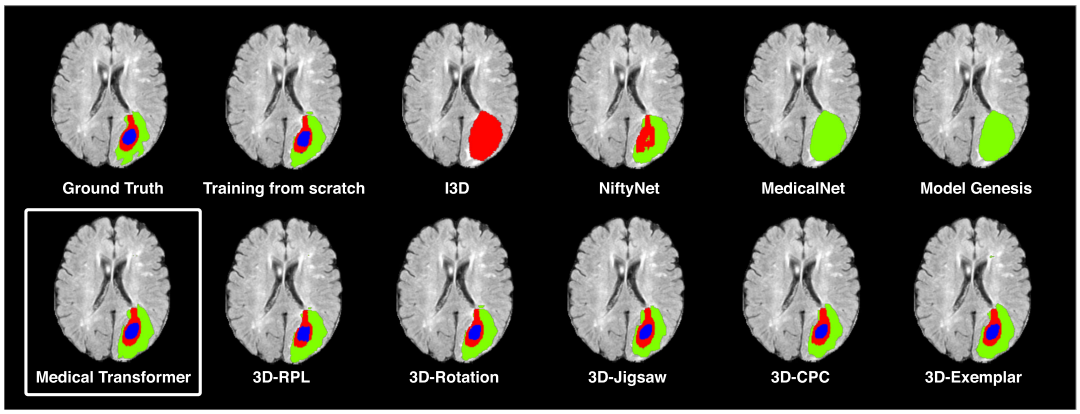


Fig. 5. Visualization of the segmentation results between our proposed Medical Transformer and the competing methods for comparison of qualitative results. The WT class includes all the visible labels (a union of green, blue, and red labels). The TC class is a union of blue and red, while the ET class is shown in red.

TABLE IV

COMPARISON OF THE NUMBER OF MODEL PARAMETERS BETWEEN COMPETITIVE METHODS AND THE PROPOSED METHOD FOR THREE TARGET TASKS

Approach	Target tasks		
	Classification	Regression	Segmentation
I3D [7]	12.378M	12.375M	17.412M
NiftyNet [8]	24.068M	24.067M	45.614M
MedicalNet [9]	33.052M	33.051M	85.798M
3D-SSL [12]	3.552M	3.551M	5.649M
Model Genesis [14]	7.093M	7.093M	19.076M
Medical Transformer	**2.402M**	**2.401M**	**2.410M**

TABLE V

COMPARISON OF THE INFERENCE TIME BETWEEN COMPETITIVE METHODS AND THE PROPOSED METHOD FOR THREE TARGET TASKS

Approach	Target tasks		
	Classification (s)	Regression (s)	Segmentation (s)
I3D [7]	0.0330	0.0326	0.0396
NiftyNet [8]	0.0746	0.0763	0.1233
MedicalNet [9]	0.0547	0.0563	0.0708
3D-SSL [12]	0.0320	0.0333	0.0609
Model Genesis [14]	0.0900	0.0889	0.1185
Medical Transformer	0.0506	0.0505	0.0730

network, which hinders the fine-tuning of the backbone network. To overcome this issue, we duplicated the weights of the pretrained input layer by the number of modalities. In terms of performance evaluation, we calculated a Dice score for each tumor region, i.e., whole tumor (WT), tumor core (TC), and enhancing TC (ET).

B. Analysis and Discussion

1) Brain Disease Diagnosis: Table II shows comparison of the results of our proposed method with those of the comparative models for AD, MCI, and NC classification in terms of mAUC. For the competing methods, we took an encoder from the pretrained model and appended a fully connected layer similar to our prediction network. Our Medical Transformer achieved the best classification performance compared with the 3-D-based SOTA methods. Compared with training from scratch, in which the model weights were randomly initialized, adopting the transfer learning scheme demonstrated clear superiority in performance by taking advantage of the universal representations and accordingly assigned better initialization weights in fine-tuning the target task model. Among the methods that were compared, I3D demonstrated a comparable performance using the recurrent layers as a feature extractor, as the classification task benefits from a high level of embedding features. MedicalNet also presented a comparable performance by pretraining the backbone network using a large-scale medical dataset.

These experimental results validated the efficacy of our multiview approach in extracting 3-D volumetric characteristics and our SSL proxy task in effectively learning universal representations. In addition, pretraining the backbone network using a large corpus MRI dataset contributed toward the superior performance of our method.

2) Brain Age Prediction: Our Medical Transformer was compared with the competing baseline methods on the brain age prediction task in Table II. It is noteworthy that our proposed method outperformed all the counterparts by a large margin. In particular, while all the competing methods showed inferior performance for this regression task, training from the scratch and Medical Transformer achieved remarkable performance with the least MAE, enhancing the effectiveness of our multiview approach to make 3-D high-level representations from three planes of a volume.

3) Brain Tumor Segmentation: As presented in Table II, it can be seen that our Medical Transformer achieved the best or comparable performance in the brain tumor segmentation task, specifically comparable performance in the evaluation of Dice WT and TC, and the best in the evaluation of Dice ET, which is the most difficult among tumor types.

The 3D-SSL methods showed consistently better performances in Dice WT and Dice TC, but a relatively poor result in Dice ET. This implies that they effectively learn 3-D structural representations through 3D SSL proxy tasks, but are inferior at capturing the sophisticated tumor regions. In contrast, I3D performed the poorest among the comparison methods, owing to a large domain gap of transfer learning between the 3-D Kinetics dataset and the medical dataset. In addition, MedicalNet showed relatively low performances in all the metrics, despite the largest number of parameters among the competing methods.

Our result of obtaining the best performance in the Dice ET evaluation showed the effectiveness of our method for

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TABLE VI

PERFORMANCE COMPARISON OF SUPERVISED AND TRANSFER LEARNING DEPENDING ON THE MULTIDIMENSIONAL APPROACHES, E.G., 2-D, 2.5-D, AND 3-D, ON BRAIN DISEASE DIAGNOSIS AND BRAIN AGE PREDICTION TASK

Approach	Supervised Learning		Transfer Learning	
	Classification	Regression	Classification	Regression
	mAUC	MAE (years)	mAUC	MAE (years)
2D ResNet	0.7336 ± 0.0186	4.5891 ± 0.3512	0.7896 ± 0.0125	3.7790 ± 0.2419
2D ResNet + Transformer	0.7389 ± 0.0098	4.5057 ± 0.6206	0.8031 ± 0.0115	3.6273 ± 0.3530
3D ResNet	0.7619 ± 0.0139	4.5181 ± 0.3355	0.8066 ± 0.0175	3.6859 ± 0.6859
3D ResNet + Transformer	0.7691 ± 0.0117	4.4771 ± 0.4299	0.8135 ± 0.0277	3.5977 ± 0.4027
2.5D (Ours)	$\mathbf{0.7728 \pm 0.0077}$	$\mathbf{4.4357 \pm 0.3260}$	$\mathbf{0.8347 \pm 0.0072}$	$\mathbf{3.4924 \pm 0.0863}$

TABLE VII

PERFORMANCE COMPARISON BETWEEN SUPERVISED LEARNING AND TRANSFER LEARNING SCHEME DEPENDING ON A RATIO OF TRAINING SAMPLES FOR THREE TARGET TASKS. THE ENTRIES IN BOLD INDICATE THE PERFORMANCE OF THE TRANSFER LEARNING SCHEME USING A PARTIAL DATASET THAT HAS REACHED THAT OF THE SUPERVISED LEARNING SCHEME USING THE ENTIRE DATASET

Scheme	Ratio	Classification	Regression	Segmentation		
		mAUC	MAE (years)	Dice WT	Dice TC	Dice ET
Supervised Learning	10%	0.5984 ± 0.0111	4.7720 ± 0.4769	0.6814 ± 0.0205	0.1417 ± 0.1668	0.1710 ± 0.1537
	30%	0.6301 ± 0.0130	4.5880 ± 0.3542	0.7921 ± 0.0125	0.5181 ± 0.0513	0.5157 ± 0.0234
	50%	0.6749 ± 0.0166	4.5291 ± 0.3377	0.8276 ± 0.0062	0.6509 ± 0.0415	0.5744 ± 0.0254
	70%	0.6936 ± 0.0075	4.5055 ± 0.3515	0.8334 ± 0.0245	0.6487 ± 0.0558	0.5641 ± 0.0355
	100%	$\mathbf{0.7728 \pm 0.0077}$	$\mathbf{4.4357 \pm 0.3260}$	$\mathbf{0.8649 \pm 0.0094}$	$\mathbf{0.6392 \pm 0.0556}$	$\mathbf{0.4830 \pm 0.0485}$
Transfer learning	10%	0.6112 ± 0.0288	$\mathbf{4.3531 \pm 0.4846}$	0.7699 ± 0.0259	0.4610 ± 0.1460	$\mathbf{0.4221 \pm 0.0906}$
	30%	0.7259 ± 0.0267	4.0142 ± 0.4403	0.8236 ± 0.0192	$\mathbf{0.6117 \pm 0.0410}$	0.5632 ± 0.0337
	50%	$\mathbf{0.7729 \pm 0.0132}$	3.7819 ± 0.2780	0.8417 ± 0.0081	0.6850 ± 0.0233	0.6216 ± 0.0110
	70%	0.7983 ± 0.0147	3.6834 ± 0.1955	0.8576 ± 0.0227	0.6769 ± 0.0610	0.5389 ± 0.0488
	100%	0.8347 ± 0.0072	3.4924 ± 0.0863	0.8733 ± 0.0086	0.6969 ± 0.0470	0.5882 ± 0.0437

capturing more sophisticated tumor regions by generating volumetric features from a combination of three plane encoding features via our multiview approach. Furthermore, the multiscale approach during fine-tuning contributed toward the performance improvement by integrating low- and high-level features.

Fig. 5 shows comparison of the qualitative results of segmentation between the competing methods and ours. We observed that the segmentation result of the proposed method was similar to that of 3D-SSL, and although it is not possible to accurately capture the border region of the WT, our proposed method was better at distinguishing between three tumor types. This result is reasonable as the transfer learning framework proposed in this work is generally applicable to a variety of downstream tasks in a parameter-efficient manner.

4) Number of Parameters: To investigate the power of our parameter-efficient Medical Transformer, we compared the number of model parameters between our method and comparable methods for three downstream tasks. As presented in Table IV, the proposed method demonstrated superior performances while having fewer parameters (92.74% fewer for classification, 92.73% fewer for regression, and 97.19% fewer for segmentation) compared with the MedicalNet, which recorded the maximum number of parameters for all the tasks.

5) Inference Time: The 2.5-D approach might reasonably be expected to substantially increase the inference time, so we compared our inference time with other 3-D-based competitive methods for three downstream tasks in Table V. The proposed method did not record the minimum inference time in all the target tasks. However, in classification, it took a time of 0.0506 s, approximately 1.5812 times that of 3D-SSL of 0.0320 s (Min) and approximately 0.5622 times that of Model Genesis of 0.0900 s (Max). In regression, it recorded 0.0505 s, approximately 1.5490 times that of I3D of 0.0326 s (Min) and approximately 0.5680 times that of Model Genesis of 0.0889 s (Max). In segmentation, it recorded 0.0730, approximately 1.8434 times that of I3D of 0.0396 s (Min) and approximately 0.5920 times that of NiftyNet of 0.1233 s (Max). These experimental results showed that our proposed method is relatively fast in inference time and computationally efficient.

6) Multidimensional Approaches (2-D/2.5-D/3-D): To represent the power of our 2.5-D approach in dealing with 3-D volumetric data, we compared the results of supervised learning and transfer learning for the proposed 2.5-D approach with those of 2-D and 3-D approaches for brain disease classification and brain age prediction in Table VI. Specifically, for 2-D and 3-D approaches, we considered: 1) 2-D ResNet; 2) 2-D ResNet + Transformer; 3) 3D ResNet; and 4) 3D ResNet + Transformer, with the network architecture parameters such as feature map size and the number of layers being set similar to the network of the proposed 2.5-D approach. Specifically, for 2-D approaches, we performed predictions by averaging slice features from a view and feeding them into the FC layer, instead of 3-D-form reconstruction. When pretraining the model for transfer learning results, given that masked encoding vector prediction through the transformer applied in our method cannot be applied in 2D and 3D ResNet, only triplet-loss-based pretraining was performed.

Our proposed method was shown to have the best results in supervised learning and transfer learning for all the tasks. For example, from the supervised learning results, mAUC was higher by 0.0037 compared with the second-best, 3D ResNet + Transformer, for classification, and MAE was lower by 0.0414 compared with the second-best, 3D ResNet + Transformer, for regression. In addition, from the transfer learning results, mAUC was higher by 0.0212 compared with the second-best, 3D ResNet + Transformer, for classification, and MAE was lower by 0.1053 compared with the second-best, 3D ResNet + Transformer, for regression. In summary, our 2.5-D method showed superior performance than 2-D and 3-D-based baselines while effectively reducing the number of parameters.

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7) Ablation Studies: First, to investigate the effect of transfer learning that performs well with a handful of training samples, we simulated experiments with fewer labeled data. Specifically, we fine-tuned the model using a partial dataset, i.e., only 10%, 30%, 50%, and 70% samples of training dataset for three target tasks. Table VII shows comparison of the results of supervised learning (training from scratch) and transfer learning for each task. Transfer learning consistently showed a significant performance improvement over the supervised learning for three tasks in all the ratio scenarios. Specifically, for brain disease diagnosis, the transfer learning scheme using only approximately 50% training data achieved similar performance to the supervised learning scheme using the entire data, thus reducing annotation efforts by nearly 50%. In addition, using only 30% of training samples, the performance for brain age prediction was better than that of training from scratch using full samples, which suggests that about 70% of the annotation cost associated with the supervised learning scheme could be saved by initializing with our Medical Transformer. In terms of brain tumor segmentation, Medical Transformer achieved similar performances to training from scratch using nearly 70% on WT, 40% on TC, and 20% on ET, where the effect of reducing annotation efforts was particularly remarkable in the evaluation of ET. Thus, these experimental results suggest that our Medical Transformer can alleviate the lack of annotated 3D MRI samples through an annotation-efficient transfer learning scheme for a variety of downstream tasks.

V. Conclusion

In this work, we proposed a novel transfer learning framework, called Medical Transformer, which effectively models 3-D volumetric images in the form of a sequence of 2-D image slices. To learn high-level volumetric representations, we took a multiview approach that leveraged plenty of information from three planes of a volume image, while providing parameter-efficient training. As a result of evaluating our pretrained model on brain disease diagnosis, brain age prediction, and brain tumor segmentation tasks, our Medical Transformer outperformed the SOTA transfer learning methods, efficiently reducing the number of parameters by up to 92% for classification and regression tasks and 97% for segmentation. Finally, it is noteworthy that our proposed framework can be applied to other types of 3-D medical images, e.g., CT, X-ray, and positron emission tomography (PET), as well as other clinical or scientific tasks.

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